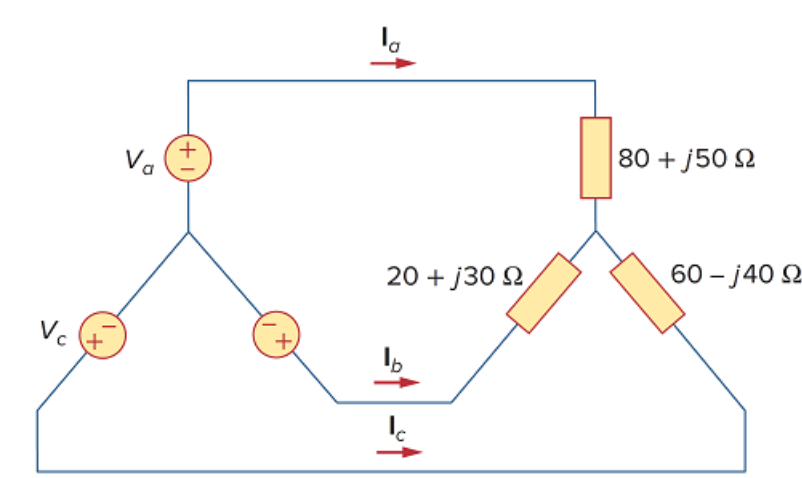


Problem

Determine the line currents for the three-phase circuit of Fig. 12.64. Let

$$\mathbf{V}_a = 220\angle 0^\circ, \mathbf{V}_b = 220\angle -120^\circ, \mathbf{V}_c = 220\angle 120^\circ \text{ V}$$

Figure 12.64



Step-by-step solution

Step 1 of 6

Refer to Figure 12.64 in the text book.

Identify the loops in the circuit diagram shown in Figure 1.

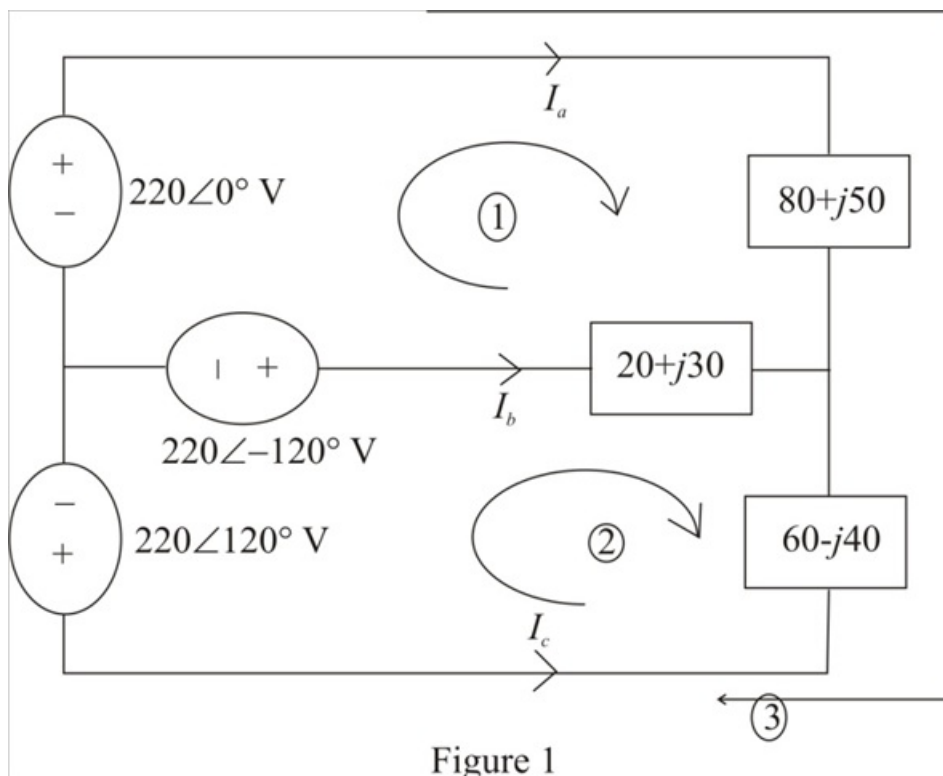


Figure 1

Step 2 of 6

Apply Kirchhoff's Voltage Law in Loop-1.

$$-220\angle 0^\circ + I_a(80 + j50) - I_b(20 + j30) + 220\angle -120^\circ = 0$$

$$I_a(94.34\angle 32^\circ) - I_b(36.06\angle 36.31^\circ) + 381.05\angle -150^\circ = 0$$

$$I_b(36.06\angle 36.31^\circ) = I_a(94.34\angle 32^\circ) + 381.05\angle -150^\circ$$

$$I_b = \frac{I_a(94.34\angle 32^\circ) + 381.05\angle -150^\circ}{36.06\angle 36.31^\circ}$$

$$I_b = I_a(2.62\angle -4.31^\circ) + 10.57\angle 173.69^\circ \dots\dots (1)$$

Therefore, the expression for the current I_b is, $I_a(2.62\angle -4.31^\circ) + 10.57\angle 173.69^\circ$.

Step 3 of 6

Apply Kirchhoff's Voltage Law in Loop-3.

$$-220\angle 0^\circ + I_a(80 + j50) - I_c(60 - j40) + 220\angle 120^\circ = 0$$

$$I_a(94.34\angle 32^\circ) - I_c(72.11\angle -33.69^\circ) + 381.05\angle 150^\circ = 0$$

$$I_c(72.11\angle -33.69^\circ) = I_a(94.34\angle 32^\circ) + 381.05\angle 150^\circ$$

$$I_c = \frac{I_a(94.34\angle 32^\circ) + 381.05\angle 150^\circ}{72.11\angle -33.69^\circ}$$

$$I_c = I_a(1.31\angle 65.69^\circ) + 5.28\angle -178.31^\circ \dots\dots (2)$$

Therefore, the expression for the current I_c is, $I_a(1.31\angle 65.69^\circ) + 5.28\angle -178.31^\circ$.

Step 4 of 6

Apply Kirchhoff's Voltage Law in Loop-2.

$$-220\angle -120^\circ + I_b(20 + j30) - I_c(60 - j40) + 220\angle 120^\circ = 0$$

$$381.05\angle 90^\circ + I_b(36.06\angle 56.31^\circ) - I_c(72.11\angle -33.69^\circ) = 0$$

Substitute $I_a(1.31\angle 65.69^\circ) + 5.28\angle -178.31^\circ$ for I_c and

$$I_a(2.62\angle -4.31^\circ) + 10.57\angle 173.69^\circ \text{ for } I_b.$$

$$-220\angle -120^\circ + I_b(20 + j30) - I_c(60 - j40) + 220\angle 120^\circ = 0$$

$$\left[381.05\angle 90^\circ + [I_a(2.62\angle -4.31^\circ) + 10.57\angle 173.69^\circ](36.06\angle 56.31^\circ) \right] - \left[I_a(1.31\angle 65.69^\circ) + 5.28\angle -178.31^\circ \right](72.11\angle -33.69^\circ) = 0$$

$$\left[381.05\angle 90^\circ + [I_a(94.48\angle 52^\circ) + 381.15\angle 130^\circ] - [I_a(94.46\angle 32^\circ) + 380.74\angle 148^\circ] \right] = 0$$

$$381.05\angle 90^\circ + I_a(94.48\angle 52^\circ) + 381.15\angle 130^\circ - I_a(94.46\angle 32^\circ) - 380.74\angle 148^\circ = 0$$

$$I_a(32.81\angle 132^\circ) + 447.67\angle 80.62^\circ = 0$$

$$I_a(32.81\angle 132^\circ) = -447.67\angle 80.62^\circ$$

$$I_a = -\frac{447.67\angle 80.62^\circ}{32.81\angle 132^\circ}$$

$$I_a = -13.64\angle -51.38^\circ$$

$$I_a = 13.64\angle 128.62^\circ \text{ A}$$

Therefore, the value of the current I_a is, $13.64\angle 128.62^\circ \text{ A}$.

Step 5 of 6

Recall equation (1).

$$I_b = I_a (2.62 \angle -4.31^\circ) + 10.57 \angle 173.69^\circ$$

Substitute $13.64 \angle 128.62^\circ$ A for I_a .

$$\begin{aligned} I_b &= (13.64 \angle 128.62^\circ)(2.62 \angle -4.31^\circ) + 10.57 \angle 173.69^\circ \\ &= 35.74 \angle 124.31^\circ + 10.57 \angle 173.69^\circ \\ &= 43.37 \angle 134.97^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_b is, $43.37 \angle 134.97^\circ$ A .

Step 6 of 6

Recall equation (2).

$$I_c = I_a (1.31 \angle 65.69^\circ) + 5.28 \angle -178.31^\circ$$

Substitute $13.64 \angle 128.62^\circ$ A for I_a .

$$\begin{aligned} I_c &= (13.64 \angle 128.62^\circ)(1.31 \angle 65.69^\circ) + 5.28 \angle -178.31^\circ \\ &= 17.87 \angle -165.69^\circ + 5.28 \angle -178.31^\circ \\ &= 23.05 \angle -168.56^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_c is, $23.05 \angle -168.56^\circ$ A .